Segmenting and Classifying Clinical Notes and Clustering Segment Labels.

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Introduction

- The goals of this project are to segment the clinical notes into different note segments and identify segments that contains similar information using classification and cluster segment labels using average segment body embeddings.
- Nurse clinical notes in the publicly available MIMIC-III data set were used in this project.
- Hierarchical Clustering and Topological Data Analysis(Mapper Algorithm) have been used for unsupervised clustering and for the classification Neural Network has been used.

Data Preprocessing

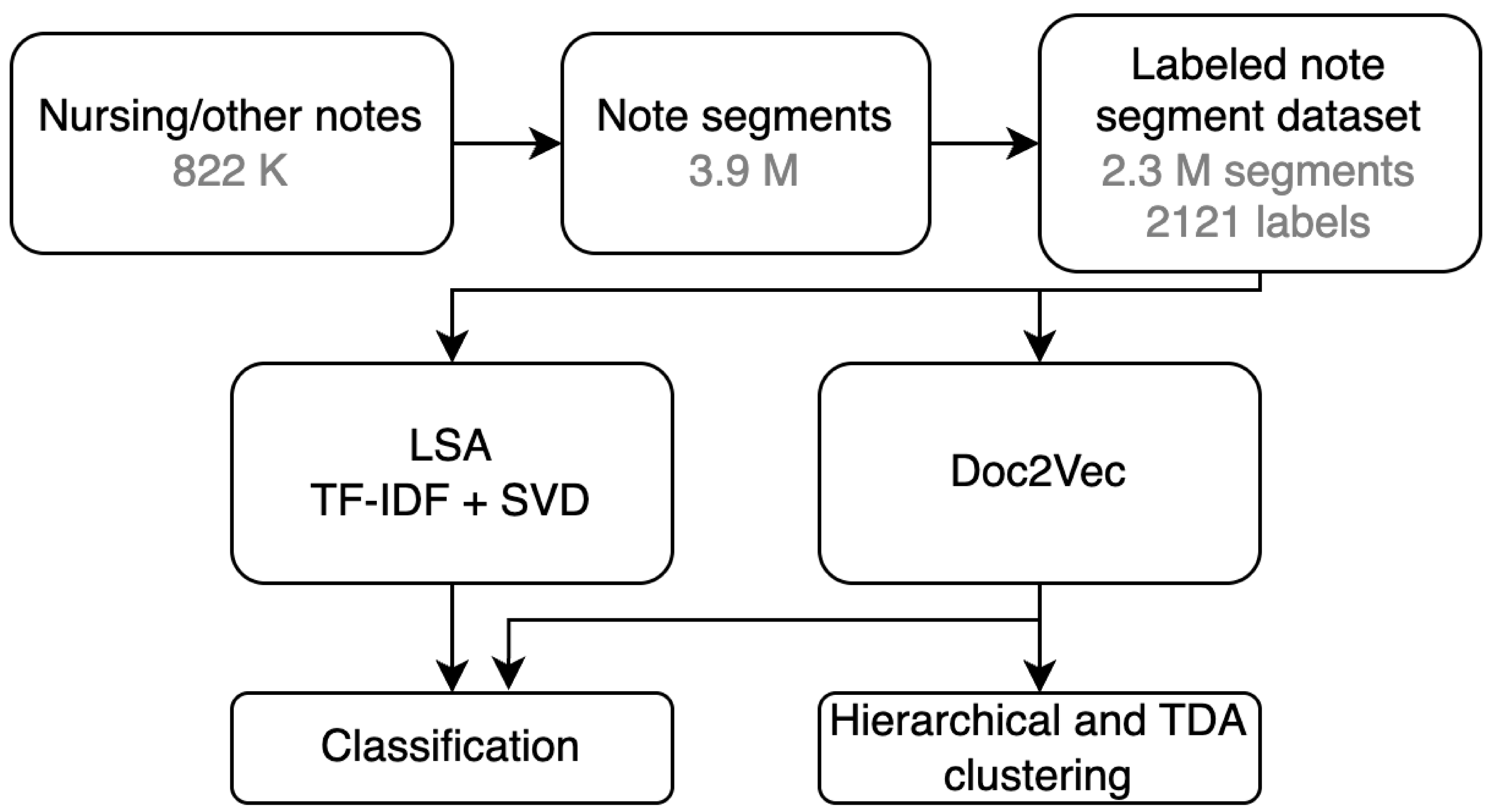
- Segmented notes into paragraph-level segments using regular expressions
- Filtered out segments with too little content or missing context.
- Normalized segment titles by removing infrequent or inconsistent labels.

Inferring Features

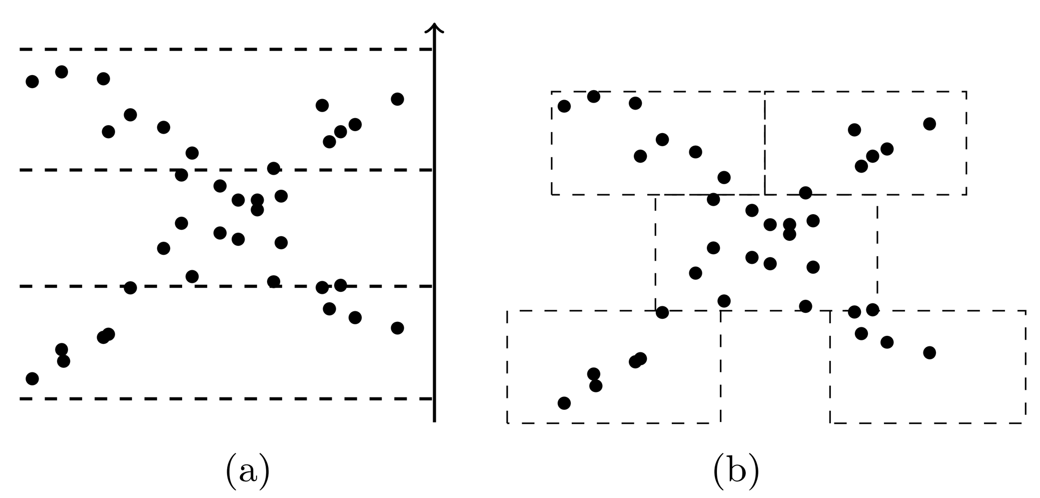
- We have used Term Frequency-Inverse Document(TF-IDF) and Doc2Vec to infer features.
- In the TF-IDF we have used Singular Value Decomposition(SVD) to reduce the dimensionality.
- Each segment is represented in low dimensional semantic space.
- In the clustering, we have used Doc2Vec to infer the features of each segment and after that we have taken average value for each segment label.

Clustering and Classification Note Segments

- Note segments were clustered using Hierarchical and TDA(Mapper) clustering methods and classified using Neural Networks.

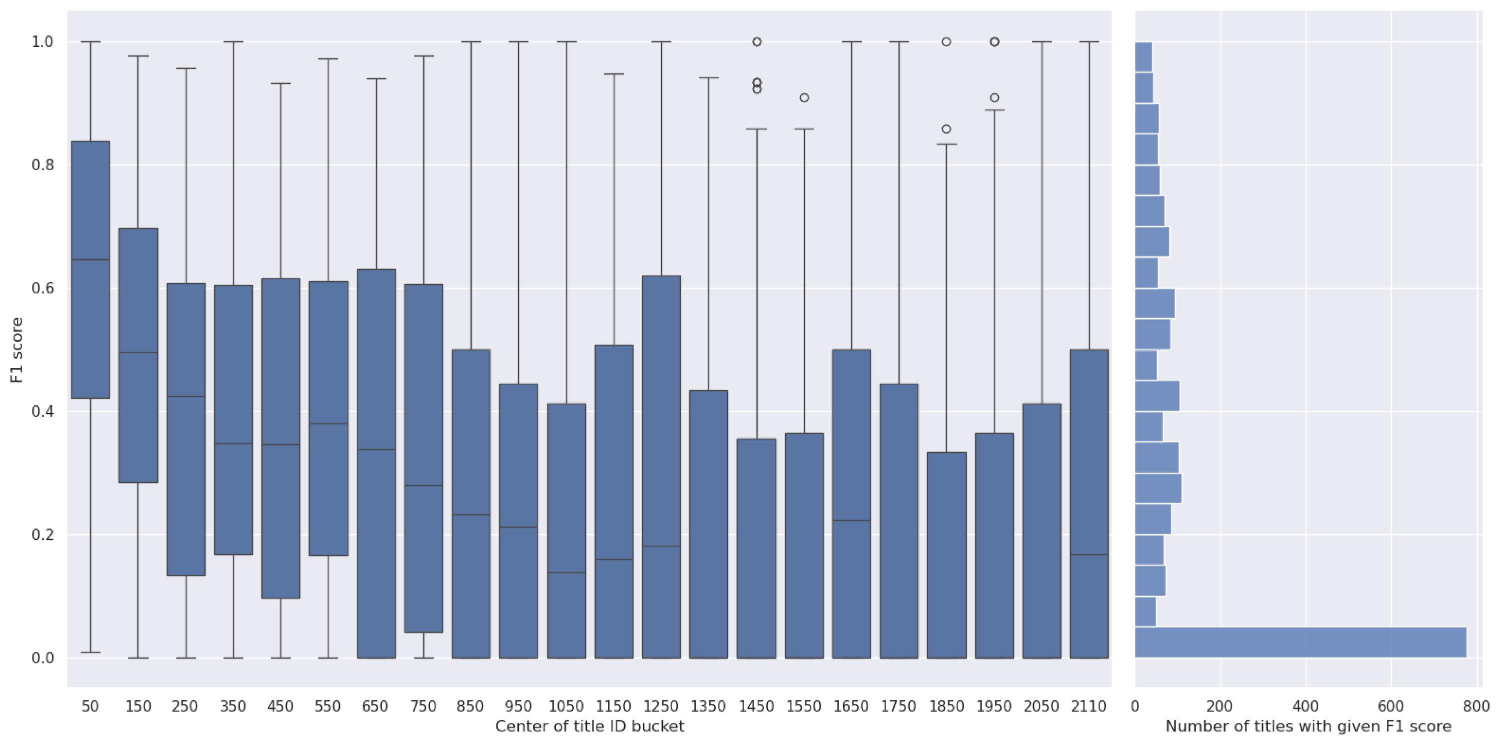


Mapper Algorithm



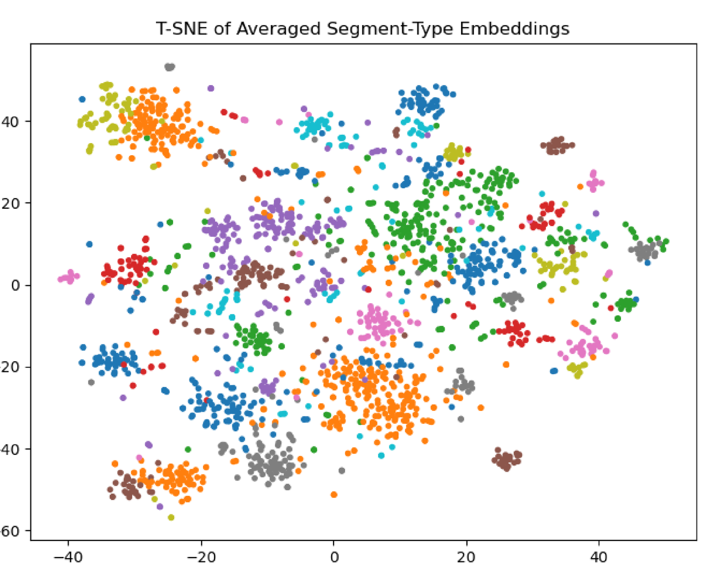
- The Mapper algorithm creates a low-dimensional representation of high-dimensional data which preserves the original data's topology.
- Data points are partitioned by a filter function (a). We create a cover of the points, and cluster within each hypercube (b).
- From each cluster we compute the nerve and contract each cluster to a point (c).

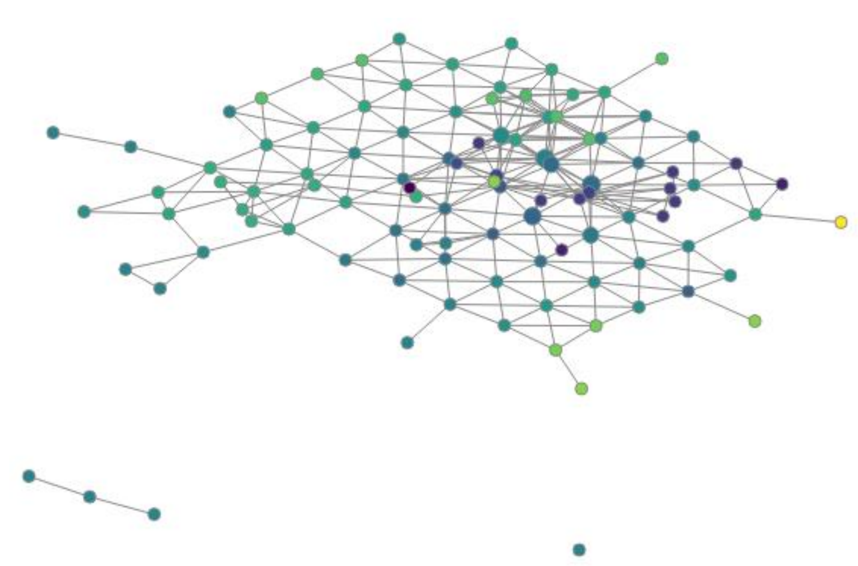
Classification Segment Bodies: Results



- Segment types are sorted by frequency (most common on the left). Each box shows F1 scores of 100 segment types grouped by frequency.
- High-frequency classes achieve higher and more stable F1 scores. Rare classes (right side) suffer from lower and more variable F1.
- Indicates that data imbalance strongly affects classification quality.
- Histogram on the right shows that most segment types have low F1.
- Motivates the need for label merging, clustering, or data augmentation for rare classes.

Clustering Segment labels: Results





- For mapper, tightly connected central regions suggest groups of labels that frequently co-occur or have semantically similar segment content.
- Peripheral or disconnected nodes likely represent rare or highly specific labels that don't overlap much with others.
- For Hierarchical clustering, We evaluated clustering quality using the silhouette score, which measures how well-separated the clusters are. Based on the score trend, we selected k = 72 to balance meaningful separation with fine-grained structure in note labels.

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